

Title

Age-Dependent Overdiagnosis Risk in Population-Level Prostate Cancer Screening: Long-Term Evidence from a Large Registry Study

Abstract

Prostate cancer detection through serological screening represents a major policy consideration given its potential to identify clinically significant disease but also to detect indolent malignancy. This analysis examined 189,386 men invited for a single PSA measurement, with 1,247 cancers detected at initial screening. Over 15 years of follow-up, cumulative prostate cancer incidence reached 7.05% in the invited group versus 6.91% in controls--an absolute excess of 0.14% (95% CI -0.05% to 0.38%). Excess case detection attributable to screening represented approximately 11.8% of initially detected cases (95% CI 0.0%-27.2%). Age-stratified competing-risk modeling demonstrated marked age dependence: among men diagnosed at age 50 years, 17% carried undetected cancer without clinical manifestation during the ensuing 15 years; this figure rose to 31% at age 70 years and 57% at age 80 years. These data suggest that age-based screening policies warrant reconsideration in healthcare settings with elevated screening penetration in older populations.